ISHAN MAHESH DESAI

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EDUCATION

University Of Illinois, Urbana-Champaign, IL, USA **August 2024 – May 2026**
Masters | Information Management, Data Science and Analytics **(GPA– 4.0/4.0)**
Relevant Coursework: Data Visualization, Data, Statistics and Information, Data Warehousing and Business Intelligence.

Dwarkadas J. Sanghvi College of Engineering, Mumbai, India **August 2020 – June 2024**
Bachelors | Information Technology *(Minors - Data Science)* **CGPA-9.08/10)**
Relevant Coursework: Big Data Analytics, Database Management System, Machine Learning, Software Engineering, Cloud Computing.

TECHNICAL SKILLS

- **Programming languages:** Python, SQL, React, HTML, CSS, R Programming
- **Frameworks & Libraries:** Pandas, NumPy, Scikit-learn, PyTorch, OpenCV, LangChain
- **Data tools:** MySQL, Tableau, Power BI, Excel, PostgreSQL, MongoDB, neo4j, DBeaver, Hive, Iceberg, PySpark
- **Application Software**: Microsoft Azure, AWS (S3, SageMaker), Atlassian, Jira, GitLab, Visual Studio, Figma

PROFESSIONAL EXPERIENCE

Data Scientist | *COUNTRY Financial, IL, USA* **May 2025 – Present**

- Built a cloud-based ML model on **AWS SageMaker** to predict car insurance claims using **Logistic Regression** and **Random Forest classifiers** resulting in **93%** accuracy and **$300K** annual claims savings.
- Developed an automated model using **Azure Document Intelligence** and **OpenAI** to extract key features from insurance documents in an **Agile environment**, eliminating **15+ hrs/week** of manual work and saving **$100K+** annually.
- Managed end-to-end **ETL pipeline** automation operations by extracting data from **Rally**, loading it into **Azure SQL Database**, and building **Tableau** visualizations to deliver insurance coverage insights for technology managers.

Technology Consultant | *Illinois Business Consulting, IL, USA* **August 2025 – Present**

- Developed 8+ **Tableau dashboards** for **Fortune 500** client's global Early Career Programs, advising on data architecture and KPI design to enable strategic workforce insights and increase **ROI by 66%.**
- Streamlined the client's global reporting process through an automated pipeline, reducing manual HR manager's effort by **50%.**
- Partnered with **cross-functional** client teams to translate KPI insights into actions that improved talent pipeline visibility.

Product Engineer | *Tata Advanced Systems Limited, Mumbai, India* **August 2023 – June 2024**

- Led a **15-member team** in developing a **facial recognition system** that ensured factory worker compliance with safety gear, detected fire and smoke, and prevented trespassing in sensitive areas.
- Orchestrated **Yolo v8, Computer Vision**, **RNN**, feature detection and extraction algorithms giving an impressive **97% precision**.
- Designed real-time dashboards for safety breach alerts, reducing violations **by 50%** and improving response time **by 35%.**

Data Analyst | *Narola Infotech, Surat, India* **August 2022 – December 2023**

- Designed 6+interactive dashboards using **Power BI** and advance **Microsoft Excel** making it convenient for stakeholders to analyze complex KPIs and sales information, **improving marketing revenue by 40%.**
- Conducted data cleaning and pre-processing using ETL techniques to the client's financial records to identify anomalies and present the information in a format conducive to seamless analysis, **improving quality by 30%**
- Detected frauds in company's financial transactions using Scikit-learn, TensorFlow, Pandas, and Matplotlib, saving **$80,000** in frauds.

Data Engineer Intern | *Servify, Mumbai, India* **July 2022 – August 2022**

- Developed a Slack integrated chatbot using **Python, Flask, SQL, and PyTorch**, enabling employees to query and retrieve sales records from backend databases via keyword-based natural language inputs, reducing manual effort by **15 hours per week.**
- Organized **SCRUM** meetings for cross-functional team coordination and managed daily tickets in **Jira** to assign tasks in **Agile environment**, prioritized development, and track project timelines for improved efficiency.

PROJECTS

AI-Powered Chatbot for Gitlab

- Designed an AI-powered knowledge assistant to retrieve information from GitLab Wikis and SharePoint, solving the issue of scattered documentation across enterprise platforms saving **300 hours+** of manual effort.
- Implemented a scalable **RAG** pipeline by integrating **Azure Cognitive Search** and **OpenAI's GPT-4**.
- Won **1st Place** at the COUNTRY Financial 2025 **Hackathon** for this innovation and later deployed the chatbot into production due to its success in reducing documentation lookup time **by over 70%,** significantly boosting team efficiency.

COVID-19 Cases in India Dashboard Analysis | Dashboard

- Built an interactive **Tableau dashboard** visualizing India COVID-19 data to uncover pandemic trends and testing disparities.
- Designed **KPI-driven visuals** for age group, vaccine dose administration, and testing capacity, enabling quick comparative insights.
- Integrated **Python preprocessing** scripts for data cleaning and aggregation, ensuring accurate analysis across multiple large datasets.

Chatbot Enhanced House Price Prediction Using Ensemble Technique | Github

- Built a real-estate prices prediction model using ensemble techniques combining SVM, Random Forest and AdaBoost Regressor.
- Incorporated 17 different parameters to make accurate prediction of properties, giving an **accuracy of 97.5 %** across various data sets.